

Mengxin Xie

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EDUCATION

Imperial College London	London, UK
MSc Risk Management & Financial Engineering	September 2023 – December 2024
▪ Core Courses: Stochastic Calculus, Financial Statistics, Advanced Option Pricing, Computational Finance with C++, Python Data Structures and Algorithms, Portfolio Management and Asset Allocation, Risk Management	
The Hong Kong Polytechnic University	Hong Kong, China
BSc (HONS) Investment Science & Finance Analytics	September 2019 – May 2023
▪ Awards: First Class Honours, Dean's Honours List	
▪ Core Courses: Machine Learning, Time Series Analysis, High-Dimensional Data Analysis, Derivative Pricing	

PROFESSIONAL EXPERIENCE

Chicago Mercantile Exchange Group Inc	London, UK
Quantitative Risk Research Analyst	July 2024 – Present
▪ SPAN2 Margin Model Research (Agriculture): Conducted empirical analysis to support the migration of agricultural products from legacy SPAN to the SPAN2 framework. Performed seasonality, correlation, and volatility diagnostics to characterize agricultural products, informing key SPAN2 margin component configurations. Analyzed Samuelson effects on historical contracts and implemented time-to-maturity scaling, improving smoothness of margin backtesting results. Enhanced margin stability through data cleansing and forward-curve extrapolation methodology improvements.	
▪ New Product Launch & Migration (Energy): Supported end-to-end energy product onboarding, including data preparation, margin backtesting, and configuration setup. Calibrated forward curves for products with limited historical data using proxy interpolation, CoC borrowing, and simulation-based methods. Collaborated with validators in production-parallel testing environments to diagnose and resolve margin discrepancies, achieving comprehensive model sign-off.	
▪ C# Library Enhancement: Extended internal C# risk libraries to support methodology enhancements; implemented bug fixes and unit tests. Investigated issues in daily curve-generation pipelines and deployed hotfixes to maintain system stability.	
Deloitte	Shanghai, China
Quantitative Modelling Intern	June 2023 – August 2023
▪ Constructed features from large-scale fundamental datasets and applied various techniques, including Synthetic Minority Over-sampling Technique (SMOTE) algorithm to address class imbalance problem.	
▪ Developed a predictive XGBoost model to assess probability of equities receiving delisting warnings on the Shanghai Stock Exchange, achieving 88.8% out-of-sample recall in identifying high-risk equities.	
Castalia Capital Management	Beijing, China
Quantitative Research Intern	June 2022 – August 2022
▪ Predicted the volatility of ETF daily returns with ARMA-GARCH model and applied volatility scaling techniques to adjust position sizes in a momentum trading strategy, improving strategy's Sharpe ratio by 13%.	
▪ Predicted stock returns using LightGBM with fundamental and technical indicators; enhanced model performance and robustness via rolling-window cross-validation and hyperparameter optimization.	

PROJECT EXPERIENCE

American Option Pricing with Monte Carlo Simulation:	
▪ Implemented Least Squares Monte Carlo, Weighted Least Squares Monte Carlo, and Parallel Monte Carlo methods to price strike-fixed American Put Options with varying initial prices, volatilities, and expiration times.	
▪ Benchmarked results against Finite Difference Method to compare algorithms' pricing speed and accuracy, and calibrated market data on the SPDR S&P 500 ETF (SPY) American put options.	
Market Regimes Identification with Clustering:	
▪ Adopted PCA and Gaussian Mixture Model clustering algorithm to identify distinct market regimes from macroeconomic indicators.	
▪ Experimented with algorithms including Linear Regression, Lasso Regression, Random Forest to select the best stock return prediction model for each regime.	
▪ Integrated best-performing models into a predictive framework to continuously monitor and rebalance an equally weighted investment portfolio from the S&P 500 stock pool monthly, achieving better backtest results than baseline model.	

ADDITIONAL SKILLS

- **Programming Skills:** Python, C++/C#, SQL
- **Language Skills:** English, Mandarin
- **Certificate:** CFA Level I (passed)